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Final Year Thesis

Master Diploma

French Title

Conception et Déploiement d'un Jumeau Numérique pour
l'Optimisation d'un Système d'Irrigation Intelligent en
Agriculture de Précision

English Title

Design and Deployment of a Digital Twin for the Optimization
of an Intelligent Irrigation System in Precision Agriculture

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Dedication

To our families, teachers, and everyone who supported us throughout this final-year project. This work is dedicated to those who believe that engineering can serve sustainability, agriculture, and society.

Acknowledgements

We would like to express our sincere gratitude to our supervisors, teachers, and the staff of École Supérieure en Informatique -08 Mai 1945- Sidi Bel Abbès for their guidance, scientific advice, and support during the realization of this project. We also thank the laboratory team who made the smart farming prototype, Libelium Smart Agriculture kit, XBee communication infrastructure, Raspberry Pi gateway, and experimental environment available for the implementation and validation of this work.

This thesis also benefited from recent research on Digital Twins, smart irrigation, precision agriculture, Internet of Things, machine learning, and sustainable agricultural water management. These works allowed us to position the proposed platform not only as a software prototype, but also as a research-driven engineering contribution.

Abstract

Agriculture is increasingly affected by water scarcity, climate variability, soil degradation, and the growing demand for food production. Irrigation management is one of the most critical factors in this context, because traditional irrigation practices are often based on fixed schedules, manual observation, or empirical experience. Such approaches can lead to water waste, plant stress, and reduced productivity. In response, this thesis presents the design and deployment of Agrio, a Digital Twin-based intelligent irrigation platform for precision agriculture.

The proposed system integrates a physical smart farming prototype, Libelium Smart Agriculture sensing equipment, XBee radio-frequency communication, a Raspberry Pi gateway, HiveMQ/MQTT messaging, a FastAPI backend, a PostgreSQL database, a Random Forest-based machine learning service, a Digital Twin state and simulation layer, a Next.js web dashboard, a Flutter mobile application, a WeMos actuator, and a water pump. The platform receives real sensor data from a controlled prototype environment, normalizes and stores the measurements, updates the virtual state of the irrigation zone, predicts irrigation need, simulates future moisture evolution, generates recommendations, supports human validation, schedules irrigation actions, and controls the physical pump through MQTT commands.

The Master 2 contribution of this work is the scientific study of Digital Twins in agriculture, with a focus on their maturity levels, enabling technologies, irrigation applications, and limitations. The engineering contribution is the implementation of an end-to-end cyber-physical prototype that connects sensing, communication, backend processing, artificial intelligence, simulation, user interfaces, deployment, and actuation. The results show that the system goes beyond a classical IoT monitoring dashboard by providing synchronization, prediction, what-if simulation, recommendation, scheduling, and physical control. The current implementation remains a prototype tested in a controlled environment, and future work should extend it to long-term field validation, stronger agronomic calibration, cybersecurity, and advanced optimization techniques.

Keywords: Digital Twin, precision agriculture, smart irrigation, IoT, MQTT, XBee, Raspberry Pi, FastAPI, PostgreSQL, machine learning, Random Forest, water

management, WeMos, smart farming.

Résumé

L'agriculture moderne fait face à des défis importants liés à la rareté de l'eau, aux variations climatiques et à la nécessité d'optimiser la production agricole. Dans ce contexte, l'irrigation intelligente constitue une application centrale de l'agriculture de précision. Ce mémoire présente la conception et le déploiement d'Agrio, une plateforme d'irrigation intelligente basée sur le concept de jumeau numérique. Le système proposé combine un prototype physique, un kit Libelium Smart Agriculture, une communication radiofréquence XBee, une passerelle Raspberry Pi, la messagerie MQTT/HiveMQ, un backend FastAPI, une base de données PostgreSQL, un service d'apprentissage automatique basé sur Random Forest, une couche de simulation du jumeau numérique, un tableau de bord web, une application mobile Flutter, un actionneur WeMos et une pompe d'eau.

La contribution scientifique porte sur l'étude de l'état de l'art des jumeaux numériques en agriculture et leur positionnement par rapport aux systèmes d'irrigation intelligents existants. La contribution d'ingénierie réside dans la mise en œuvre d'une chaîne complète reliant les capteurs, la communication IoT, le traitement backend, la prédiction, la simulation, la recommandation, la validation humaine et le contrôle physique de l'irrigation. Les résultats montrent que le système dépasse la simple visualisation IoT en intégrant la synchronisation, la prédiction, la simulation et l'action.

Mots-clés : jumeau numérique, agriculture de précision, irrigation intelligente, IoT, MQTT, apprentissage automatique, FastAPI, PostgreSQL.

ملخص

تواجه الزراعة الحديثة تحديات كبيرة مرتبطة بندرة المياه والتغيرات المناخية والحاجة إلى تحسين الإنتاجية. في هذا السياق، تمثل أنظمة الري الذكي أحد أهم تطبيقات الزراعة الدقيقة. يقدم هذا العمل منصة **Agrio**، وهي منصة ري ذكية تعتمد على مفهوم التوأم الرقمي، حيث تجمع بين نموذج فيزيائي حقيقي، حساسات زراعية، اتصال لاسلكي، بوابة **Pi**، **Raspberry** بروتوكول **MQTT**، خادم خلفي **FastAPI**، قاعدة بيانات **PostgreSQL**، خدمة تعلم آلي، محاكاة رقمية، واجهة ويب، تطبيق هاتف، ومتحكم **WeMos** للتحكم في مضخة الماء.

تتمثل المساهمة العلمية في دراسة حالة الفن للتوائم الرقمية في الزراعة ومقارنة الأعمال الحالية، بينما تتمثل المساهمة الهندسية في بناء سلسلة كاملة من جمع البيانات إلى التنبؤ والمحاكاة والتوصية والتحكم الفعلي في المضخة. يبين العمل أن النظام لا يقتصر على مراقبة البيانات، بل يوفر تمثيلاً رقمياً متزامناً ودعماً ذكياً لاتخاذ القرار.

List of Abbreviations

Table 1: Abbreviations used in the thesis

Abbreviation	Meaning
AI	Artificial Intelligence
ADT	Agricultural Digital Twin
API	Application Programming Interface
CIS	Computer Systems Engineering
AIDS	Artificial Intelligence and Data Science
CPS	Cyber-Physical System
DB	Database
DSS	Decision Support System
DT	Digital Twin
ET	Evapotranspiration
ETo	Reference Evapotranspiration
FAO	Food and Agriculture Organization
FMIS	Farm Management Information System
IoT	Internet of Things
JSON	JavaScript Object Notation
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
REST	Representational State Transfer
RF	Radio Frequency
RDI	Regulated Deficit Irrigation
RFID	Radio-Frequency Identification
SBA	Sidi Bel Abbès
SQL	Structured Query Language
SWB	Soil Water Balance

Abbreviation	Meaning
UAV	Unmanned Aerial Vehicle
UI	User Interface
WSN	Wireless Sensor Network

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General Introduction

Context and Problem Statement

Agriculture is one of the foundations of food security, economic development, and social stability. However, it is also one of the sectors most exposed to climate variability, water scarcity, soil degradation, and increasing production pressure. In many regions, irrigation remains a central factor determining crop quality and yield. When irrigation is applied too late, plants may suffer water stress and productivity may decrease. When irrigation is applied too early or in excessive quantities, water is wasted, soil nutrients may leach, and the cost of operation increases. The challenge is therefore not simply to irrigate, but to irrigate at the right time, with the right quantity, and according to the actual state of the soil, crop, and environment.

Traditional irrigation practices are often based on fixed schedules, manual field observation, or the farmer's experience. These practices remain valuable because farmers possess contextual knowledge of their crops and land. Nevertheless, they can become insufficient when environmental conditions change quickly or when water availability becomes uncertain. Precision agriculture responds to this issue by relying on sensors, communication systems, data processing, and intelligent decision support. Smart irrigation is one of the most important applications of precision agriculture because it directly addresses water-use efficiency and sustainability.

Digital Twin technology offers a more integrated approach. A Digital Twin is not only a dashboard or a static simulation. It is a virtual representation of a physical system, continuously updated using real data, able to simulate future behavior, support decision-making, and potentially influence the physical system through feedback or actuation. In agriculture, Digital Twins can represent farms, plots, irrigation zones, greenhouses, plants, soil profiles, livestock, or agricultural machinery. Their value lies in connecting physical measurements, computational models, prediction algorithms, and operational decisions within a coherent cyber-physical loop.

This Master Diploma report focuses on the scientific part of Agrio. Its purpose is to study the foundations of precision agriculture and smart irrigation, define the concepts and maturity levels of Digital Twins, review the state of the art of agricultural

Digital Twins and intelligent irrigation, and synthesize the main gaps that justify the proposed Agrio direction.

Objectives

The central research question of this Master report is: how can the literature on Digital Twins, IoT-based smart irrigation, and machine learning irrigation systems be analyzed in order to position Agrio scientifically? This question is divided into secondary questions: what are the main concepts and maturity levels of Digital Twins in agriculture? Which technologies are commonly used in IoT-based smart irrigation systems? How can machine learning support irrigation decision-making? What are the strengths and limitations of current research? Where does the proposed Agrio prototype stand compared with existing works?

The objectives of the Master Diploma report are: to study Digital Twins and smart irrigation systems in the agricultural domain; to present the fundamental concepts of precision agriculture, IoT, artificial intelligence, and simulation; to review existing research on agricultural Digital Twins and irrigation systems; to compare related works according to clear scientific criteria; and to identify the novelty, limitations, and positioning of Agrio.

Thesis Outline

The report is organized according to the Master Diploma structure. The front matter contains the acknowledgements, abstracts, and lists. The introduction presents the context, problem statement, objectives, and outline. The background part introduces precision agriculture, smart irrigation, Digital Twins, IoT, and artificial intelligence. The state-of-the-art part reviews Digital Twins in agriculture and irrigation. The synthesis part compares existing works and positions Agrio scientifically. The report ends with a general conclusion.

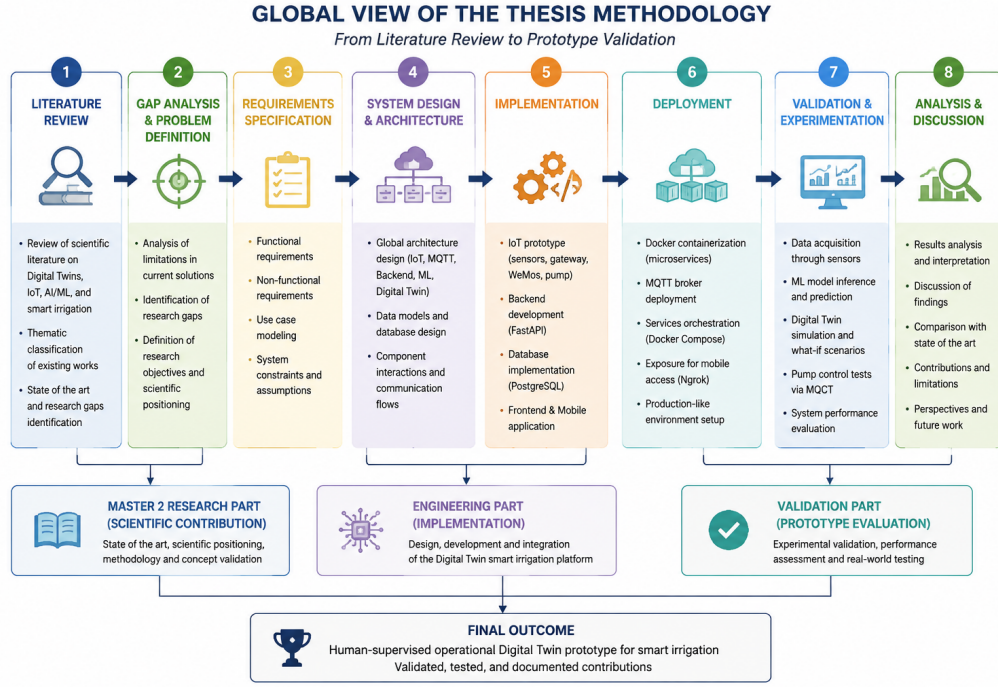


Figure 1: Global view of the thesis methodology from literature review to prototype validation. Source: Authors.

Table 2: Summary of Master Diploma objectives and contributions

Dimension	Objective	Contribution
Scientific background	Study Digital Twins, smart irrigation, IoT, and ML concepts	Structured theoretical foundation
State of the art	Analyze recent research on agricultural Digital Twins and smart irrigation	Literature review and research gap identification
Synthesis	Compare existing systems according to Digital Twin maturity and irrigation functionality	Scientific positioning of Agrio

Part I

Theoretical Background

Chapter 1

Precision Agriculture and Smart Irrigation

1.1 Introduction

Precision agriculture is a management approach that uses data and technology to adapt agricultural practices to spatial and temporal variability. Instead of treating a field as a uniform entity, precision agriculture recognizes that soil properties, water availability, crop development, microclimate, pest pressure, and nutrient status can differ across zones and over time. This approach is especially important for irrigation, where uniform schedules may over-irrigate some areas and under-irrigate others.

Smart irrigation is the application of sensing, communication, computation, and control to irrigation management. It aims to apply water according to actual crop and soil needs. In the literature, smart irrigation systems are often described as combinations of sensors, wireless networks, cloud platforms, control units, data processing, and user interfaces. García et al. [7] review sensors, nodes, wireless technologies, and irrigation actuators used in IoT-based irrigation systems, while Obaideen et al. [8] emphasize the role of soil and weather monitoring in achieving water-use efficiency.

In this thesis, smart irrigation is understood as an operational system that can measure relevant variables, transmit them to a processing layer, infer the irrigation need, and support or execute irrigation decisions. The proposed Agrio platform follows this definition but extends it through a Digital Twin layer that keeps a virtual state of the irrigation zone and simulates future moisture evolution.

CONCEPTUAL SCOPE OF PRECISION AGRICULTURE AND SMART IRRIGATION

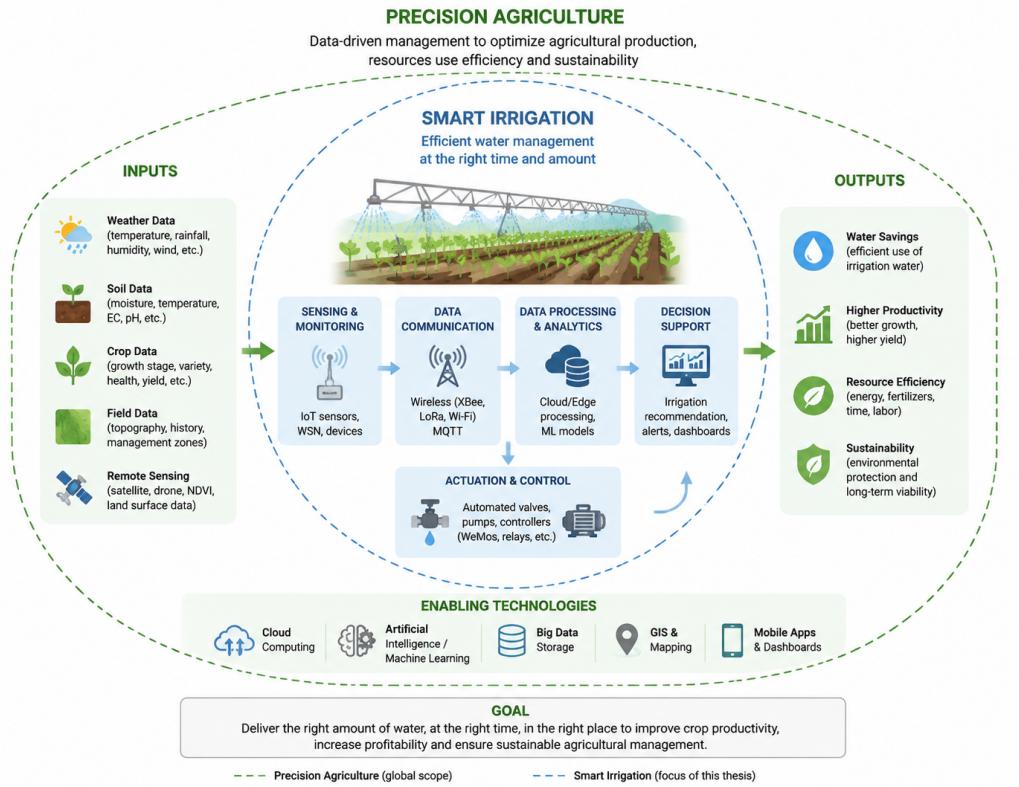


Figure 1.1: Conceptual scope of precision agriculture and smart irrigation. Source: Authors.

1.2 Water scarcity and irrigation management

Water scarcity is one of the main motivations behind intelligent irrigation research. Several reviewed papers remind that agriculture accounts for a large share of global freshwater withdrawals, often around 70%. This fact is used in many irrigation and Digital Twin studies to justify the need for more efficient water management. The issue is not only environmental; it is also economic and social. Water scarcity can reduce yields, increase food prices, and make agricultural production more vulnerable to drought and climate variability.

Irrigation scheduling is the process of deciding when and how much water should be applied. Classical scheduling may rely on fixed calendars, farmer experience, empirical observation of soil moisture, crop coefficients, or soil water balance methods. These methods can be effective when they are correctly calibrated, but they may become inaccurate when weather, soil, crop stage, or irrigation infrastructure changes. In real agricultural environments, local conditions can cause deviations between estimated and actual crop water consumption.

In the context of this project, the water management problem is addressed at

prototype scale. The objective is not to claim immediate large-scale water saving, but to design and validate a technical architecture capable of supporting precise irrigation decisions. This distinction is important academically: the prototype proves the feasibility of the Digital Twin workflow, while future field trials are required to quantify agronomic and economic water savings under real farm conditions.

1.3 From conventional irrigation to smart irrigation

Conventional irrigation methods include flood irrigation, furrow irrigation, sprinkler irrigation, drip irrigation, and manual watering. Each method has advantages and constraints depending on crop type, soil texture, topography, water availability, and farm infrastructure. Drip irrigation is often considered efficient because it delivers water close to the root zone, but even drip systems can waste water if scheduling is poorly adjusted.

IoT-based smart irrigation introduces sensing and automation into this process. Soil moisture sensors can indicate whether the root zone contains sufficient water. Temperature and humidity sensors help estimate environmental demand. Rainfall sensors and weather forecasts help avoid unnecessary irrigation before or after rainfall. Wind and radiation information can support evapotranspiration estimation. Actuators such as pumps, valves, and relays execute irrigation actions.

García et al. [7] show that many IoT irrigation systems use pumps or valves, soil moisture sensors, microcontrollers, and wireless communication. Obaideen et al. [8] explain that smart irrigation includes data acquisition, irrigation control, wireless communication, data processing, and fault detection. These components are also present in Agrio, but the project adds Digital Twin state management, machine learning prediction, scenario simulation, and user validation.

Table 1.1: Evolution from conventional irrigation to Digital Twin irrigation

Approach	Decision basis	Strength	Limitation
Manual irrigation	Farmer observation and experience	Simple and context-aware	Difficult to scale and may miss rapid changes
Scheduled irrigation	Calendar and fixed duration	Easy to implement	Insensitive to actual soil moisture and weather
Sensor-triggered irrigation	Thresholds on soil moisture or weather	Responsive to measured data	Usually limited to current state and simple rules

Approach	Decision basis	Strength	Limitation
ML-assisted irrigation	Historical and real-time features	Learns complex patterns	Requires data quality, model validation, and interpretability
Digital Twin irrigation	Synchronized virtual state, prediction, simulation, feedback	Supports what-if analysis and decision traceability	Requires integration, calibration, and cyber-physical reliability

1.4 Main variables in smart irrigation

The effectiveness of a smart irrigation system depends on the variables it monitors. Soil moisture is the central variable because it indicates water availability in the root zone. Soil temperature influences root activity and water uptake. Air temperature, humidity, wind speed, and light intensity influence evapotranspiration. Rainfall indicators determine whether natural precipitation can replace irrigation. Leaf wetness can support crop health monitoring and disease risk assessment.

The Libelium Smart Agriculture kit used in this project supports a wide range of environmental and agricultural measurements. The MQTT payloads generated by the system include soil moisture, soil temperature, air temperature, humidity, pressure, leaf wetness, light intensity, rainfall current/hour/day, wind direction, wind speed, timestamp, and device identifier. This multi-variable sensing approach is important because irrigation decisions should not be based on soil moisture alone.

However, more variables also imply more processing complexity. Sensor values may arrive in different formats, may contain missing values, or may require normalization. For example, soil moisture may be sent as a fraction and converted to percentage; light intensity may include suffixes that must be parsed; wind direction may be categorical and converted into degrees or cyclic features. These issues are handled by the backend parser and dataset adaptation pipeline.

Table 1.2: Measured variables and their role in Agrio

Variable	Role in irrigation decision	Used in Agrio
Soil moisture	Primary indicator of water availability	Yes
Soil temperature	Influences root activity and soil water dynamics	Yes
Air temperature	Supports evapotranspiration and stress estimation	Yes

Variable	Role in irrigation decision	Used in Agrio
Humidity	Indicates atmospheric demand and evaporation conditions	Yes
Rainfall	Avoids unnecessary irrigation and supports forecast decisions	Yes
Wind speed/direction	Influences evapotranspiration and environmental interpretation	Yes
Light intensity	Supports environmental context and crop-energy conditions	Yes
Leaf wetness	Supports plant/environment monitoring	Yes
Pressure	Additional weather/environment indicator	Yes
Pump status	Closes the loop between decision and actuation	Yes

1.5 Communication technologies in smart irrigation

Smart irrigation requires reliable communication between sensors, gateways, cloud services, and actuators. The literature reports many possible communication technologies: Wi-Fi, Bluetooth, Zigbee, XBee, LoRa, LoRaWAN, GSM/GPRS, NB-IoT, and cellular networks. The best choice depends on field size, power constraints, range, data rate, cost, and infrastructure availability.

In Agrio, the field-side communication uses XBee radio-frequency communication. The XBee gateway receives the transmitted sensor data, and a Raspberry Pi processes the received frames. The gateway then publishes a standardized JSON payload to HiveMQ using MQTT. MQTT is well adapted to IoT communication because it follows a publish/subscribe model and supports decoupling between data producers and consumers. The backend subscribes to sensor topics, while the WeMos actuator subscribes to command topics and publishes status feedback.

This layered communication approach is important because it separates the field hardware protocol from the cloud/backend protocol. The backend does not need

to interpret raw radio frames; it only consumes normalized MQTT payloads. This makes the architecture more modular and easier to extend to other sensor stations or communication technologies in the future.

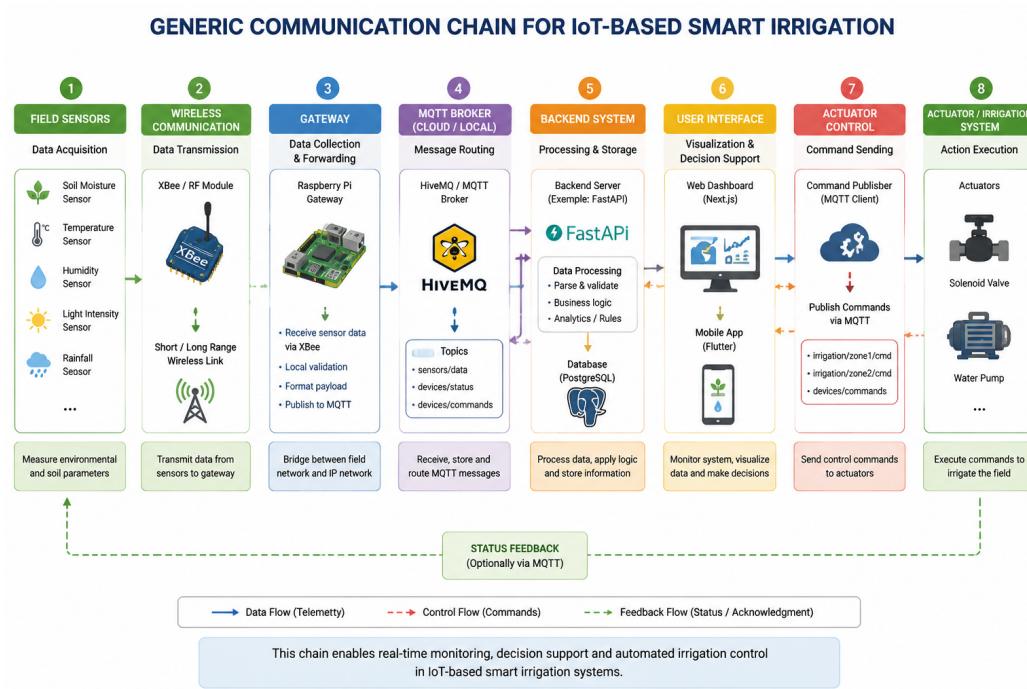


Figure 1.2: Generic communication chain for IoT-based smart irrigation. Source: Authors.

1.6 Actuators and irrigation control

Actuators transform digital decisions into physical irrigation actions. In irrigation systems, common actuators include pumps, solenoid valves, relays, fertigation injectors, and motorized valves. In a smart irrigation architecture, actuation must be carefully controlled because an incorrect command can waste water or damage the prototype. Therefore, the system must distinguish between recommendation, validation, scheduling, and physical execution.

The Agrio prototype uses a WeMos device to control the pump through digital pins. In the current implementation, D5 is used for pump control and D3 is used for an auxiliary lamp. The backend publishes MQTT commands such as `ON_D5` and `OFF_D5` to the WeMos command topic. The WeMos publishes status messages indicating the state of D5 and D3. The dashboard reads the status through the backend and allows manual control or validated scheduled operation.

The system intentionally separates simulation from actuation. Running a Digital Twin simulation does not start the pump. Simulation only predicts a virtual outcome. Physical actuation occurs through explicit manual command or scheduled

execution after validation. This design choice improves safety and makes the prototype more acceptable for academic demonstration.

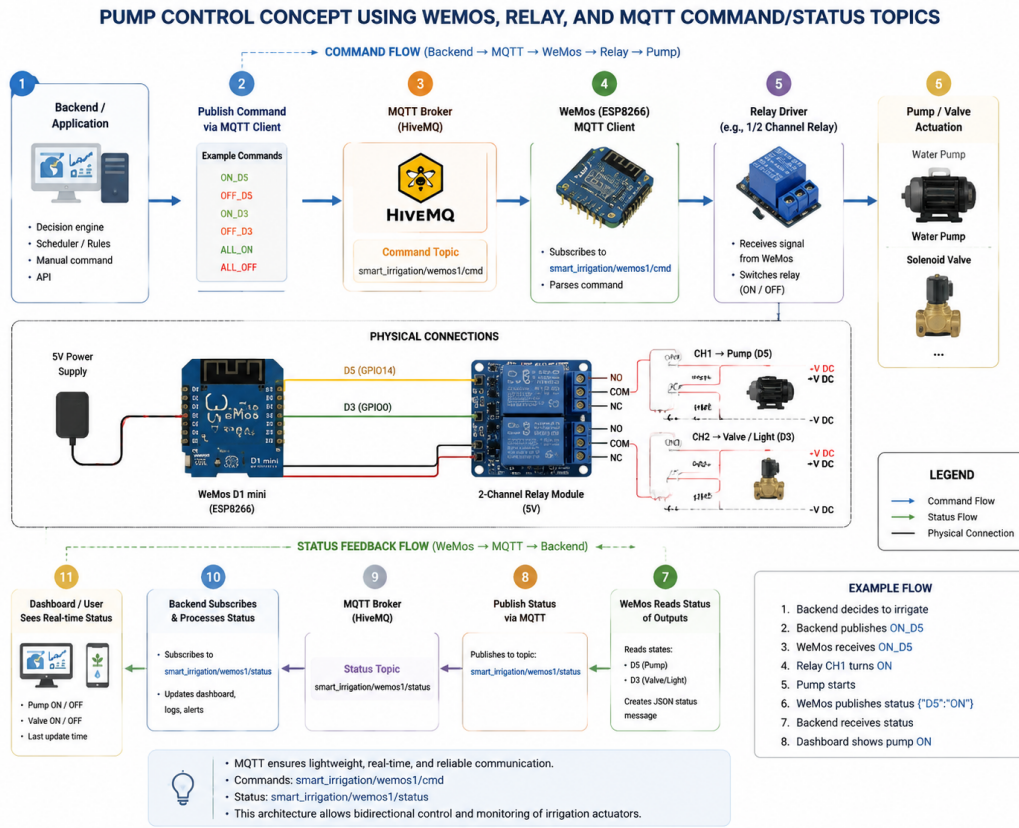


Figure 1.3: Pump control concept using WeMos, relay, and MQTT command/status topics. Source: Authors.

1.7 Challenges of smart irrigation systems

The literature identifies several challenges that limit the adoption of smart irrigation: high sensor cost, sensor calibration, communication reliability, power management, data security, interoperability, model generalization, farmer acceptance, and limited long-term validation. The agricultural environment is also more variable than many industrial environments. Soil heterogeneity, crop growth, weather uncertainty, and biological responses make prediction and control difficult.

Many existing systems demonstrate promising results in laboratory or small-scale settings but lack extended field testing. This limitation is also relevant for Agrio. The current prototype demonstrates the complete technical workflow in a controlled environment, but long-term agronomic validation in real farm conditions remains future work. A rigorous academic thesis should therefore present both the strengths and limitations of the project transparently.

Another challenge is the gap between decision support and autonomous control.

Fully automatic irrigation may be attractive, but it requires strong safety, calibration, and trust. Human-in-the-loop validation is a practical compromise for a final-year prototype: the system can recommend and simulate, while the user remains responsible for confirming or scheduling physical irrigation.

1.8 Conclusion

This chapter presented the context of precision agriculture and smart irrigation. It showed that irrigation optimization requires sensing, communication, data processing, decision support, and actuator control. The chapter also introduced the importance of soil moisture, weather variables, wireless communication, and safe pump control. These concepts prepare the transition toward Digital Twin technology, which is presented in the next chapter as the conceptual foundation of Agrio.

Chapter 2

Digital Twins, IoT, and Artificial Intelligence

2.1 Introduction

Digital Twin technology has become a key concept in Industry 4.0 and is increasingly studied in agriculture. The term refers to a virtual counterpart of a physical asset, process, or system. Unlike a static digital model, a Digital Twin is connected to the physical system through data flows and can be updated according to the real state of the asset. In advanced cases, the Digital Twin can also influence the physical system through recommendations or actuation.

Agriculture is a challenging domain for Digital Twins because it includes living organisms, heterogeneous soils, uncertain weather, spatial variability, and economic constraints. Nevertheless, recent reviews agree that Digital Twins can improve monitoring, prediction, decision-making, resource optimization, and sustainability in smart farming [1]-[6]. This thesis focuses specifically on Digital Twins for intelligent irrigation.

2.2 Digital Model, Digital Shadow, and Digital Twin

A useful distinction in the literature is the separation between Digital Model, Digital Shadow, and Digital Twin. A Digital Model is a static or offline representation of a physical system. It may simulate crop growth or water balance using historical or manually entered data, but it is not automatically updated by the physical system. A Digital Shadow introduces a one-way data flow from the physical system to the virtual system. Sensor data can update the digital representation, but the virtual system

does not directly influence the physical system. A Digital Twin adds bidirectional interaction: the physical system updates the virtual model, and the virtual model can support or trigger decisions that affect the physical system.

This distinction is important for positioning Agrio. The system is more than a Digital Model because it receives real MQTT data from the physical prototype. It is also more advanced than a simple Digital Shadow because it supports recommendation, validation, scheduling, and pump control. However, because full autonomous control is intentionally restricted for safety, Agrio can be described as a human-supervised operational Digital Twin prototype rather than a fully autonomous industrial Digital Twin.

Figure 2.1 should show the three maturity levels. It can be adapted conceptually from the literature, but it is preferable to redraw it in an original style for the thesis to avoid copyright issues.

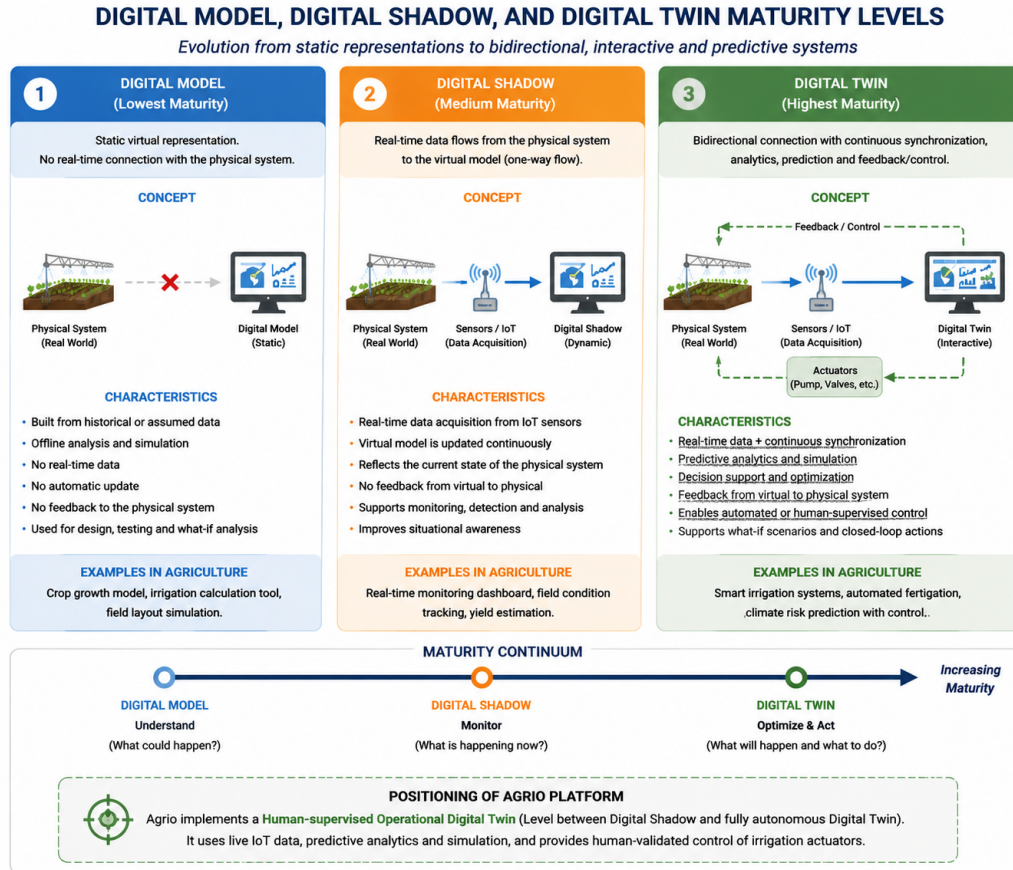


Figure 2.1: Digital Model, Digital Shadow, and Digital Twin maturity levels. Source: Authors.

2.3 Digital Twin architecture layers

Most Digital Twin architectures include a physical layer, sensing layer, communication layer, data management layer, modeling and simulation layer, analytics or intelligence layer, application layer, and feedback/control layer. In agriculture, the physical layer may include crops, soil, irrigation equipment, greenhouse systems, livestock, or machinery. The sensing layer collects data from sensors, cameras, satellites, drones, or weather stations. The communication layer transfers the data using wireless or wired protocols. The data layer stores and organizes the information. The modeling layer builds the virtual representation. The intelligence layer predicts, optimizes, or recommends. The application layer exposes the information to users. The feedback layer sends instructions to actuators or human operators.

Agrio implements these layers through concrete technologies. The physical layer is the controlled plant prototype with irrigation pipes and pump. The sensing layer is the Libelium Smart Agriculture kit. The communication layer includes XBee, Raspberry Pi gateway processing, and HiveMQ/MQTT. The data layer uses PostgreSQL. The backend and Digital Twin services create zone states and histories. The ML service predicts irrigation need and suggested hour. The dashboard and mobile application present information to the user. The WeMos and pump represent the actuator layer.

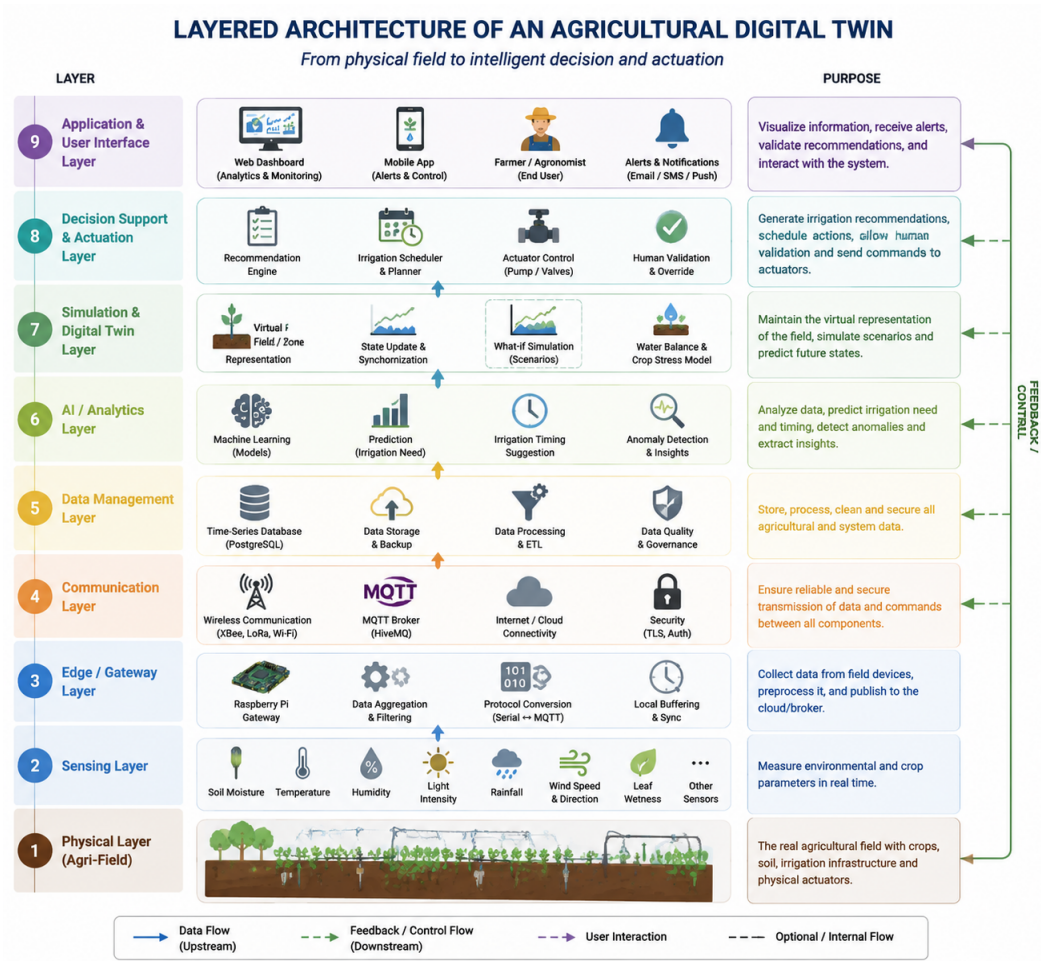


Figure 2.2: Layered architecture of an agricultural Digital Twin. Source: Authors.

2.4 IoT as the physical-to-digital bridge

IoT technologies are essential to Digital Twins because they provide the connection between physical assets and virtual models. Without live or near-real-time data, a Digital Twin remains a simulation or a static model. In smart irrigation, IoT devices collect soil and environmental data, transmit them to cloud or edge services, and support remote control of pumps or valves.

MQTT is commonly used in IoT systems because it follows a publish/subscribe communication pattern. Devices publish messages to topics, and subscribers receive messages from those topics. This decoupling is useful in Agrio because the gateway, backend, dashboard, and WeMos actuator do not need to communicate through direct synchronous calls. Instead, they exchange information through topics such as sensors/data, smart_irrigation/wemos1/status, and smart_irrigation/wemos1/cmd.

2.5 Artificial intelligence and machine learning in irrigation

Machine learning supports irrigation decision-making by learning patterns from historical and real-time data. Models can estimate soil moisture, predict evapotranspiration, classify irrigation need, suggest irrigation timing, or optimize water amount. Del-Coco et al. [9] describe smart irrigation as a combination of sensors, IoT technologies, algorithms, datasets, and decision layers. They also emphasize that ML-based smart irrigation requires reliable datasets and careful evaluation.

Common algorithms include Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting, Artificial Neural Networks, LSTM networks, and hybrid models. Random Forest is attractive for engineering prototypes because it is robust, handles nonlinear relationships, can work with mixed feature sets, and is less difficult to train than deep learning models. In Agrio, Random Forest models are used for two tasks: predicting whether irrigation is needed and suggesting an irrigation hour.

The thesis must be careful when presenting ML results. The metadata of the trained models can report high accuracy, but these results are training/evaluation metrics on the available dataset, not guaranteed field performance. Real agronomic validation would require long-term field experiments under different crops, soils, and weather conditions.

Table 2.1: Machine learning algorithms used in smart irrigation literature

Algorithm	Typical role	Advantages	Limitations
Decision Tree	Irrigation/no-irrigation rules and duration decisions	Interpretable, low computation cost	Can overfit simple datasets
Random Forest	Classification of irrigation need and robust tabular prediction	Robust, nonlinear, suitable for mixed variables	Less transparent than a single tree
SVM	Classification/regression for irrigation status	Effective in smaller datasets	Requires tuning and scaling
Gradient Boosting	Soil moisture and evapotranspiration prediction	High predictive performance	More complex tuning
ANN/LSTM	Time-series and nonlinear water demand prediction	Can model temporal dynamics	Requires larger datasets and careful validation
Reinforcement Learning	Optimization of irrigation policy over time	Can learn sequential decisions	Needs simulation environment and safety constraints

2.6 Simulation and what-if analysis

Simulation is a core capability of Digital Twins. In irrigation, what-if simulation can compare moisture evolution if irrigation is applied versus if it is skipped. This enables the user to evaluate a decision before applying it physically. Simulation can also help explain AI decisions because the user can see the expected trajectory rather than only a binary recommendation.

Agrio implements a Digital Twin simulation workspace. The user selects a zone, reads the live state, sets scenario inputs such as irrigation duration and projection horizon, runs the simulation, and compares outputs. The system displays predicted moisture, water balance, 24-hour and 48-hour projections, and a textual explanation. The simulation does not activate the pump; it only supports decision-making.

2.7 Human-in-the-loop Digital Twins

Human-in-the-loop design is important in agricultural Digital Twins. Farmers and agronomists possess domain knowledge that cannot be fully replaced by algorithms, especially in early prototypes. Human validation also improves safety when actuators are involved. In Agrio, recommendations can be validated by the user before becoming scheduled irrigation events. Manual pump control remains possible through the dashboard.

This design supports trust. Instead of hiding the logic behind automation, the platform shows recommendation status, confidence, duration, suggested time, and explanations. The user can then decide whether to accept the system output. In future versions, the system could learn from user feedback and execution results.

2.8 Digital Twin challenges in agriculture

Digital Twin adoption in agriculture faces major barriers: data heterogeneity, sensor calibration, limited connectivity, lack of standard architectures, high implementation cost, privacy and security concerns, limited interoperability, and difficulty of modeling biological systems. Recent reviews also note that many agricultural systems use the term Digital Twin loosely, even when they are closer to decision support tools or monitoring dashboards [5].

Agrio addresses some of these challenges by defining a complete technical architecture and by separating sensing, communication, backend processing, ML, simulation, user validation, and actuation. However, the project does not solve all challenges. It still requires long-term field validation, improved security, stronger agronomic cal-

ibration, and extension to multiple crops and zones.

2.9 Conclusion

This chapter presented the theoretical foundations of Digital Twins, IoT, and AI for smart irrigation. It clarified the difference between digital models, digital shadows, and Digital Twins, introduced the layered architecture required for cyber-physical synchronization, and explained the role of ML and what-if simulation in decision support. These concepts are used in the next chapters to analyze the state of the art and position the Agrio project.

Part II

State of the Art

Chapter 3

Digital Twins in Agriculture and Irrigation: State of the Art

3.1 Review methodology

The state-of-the-art study was built from peer-reviewed papers and technical sources related to agricultural Digital Twins, smart irrigation, IoT-based monitoring, machine learning for irrigation, and Digital Twin water management. The reviewed papers include general reviews, systematic reviews, bibliometric reviews, and application papers. The objective was not only to summarize publications, but to identify patterns, limitations, and opportunities relevant to the proposed Agrio platform.

The review was organized around five questions: (1) How are Digital Twins defined and classified in agriculture? (2) Which enabling technologies are used? (3) How are Digital Twins applied to irrigation and water management? (4) Which machine learning approaches support irrigation decisions? (5) What gaps remain in current systems, and how does Agrio address some of them?

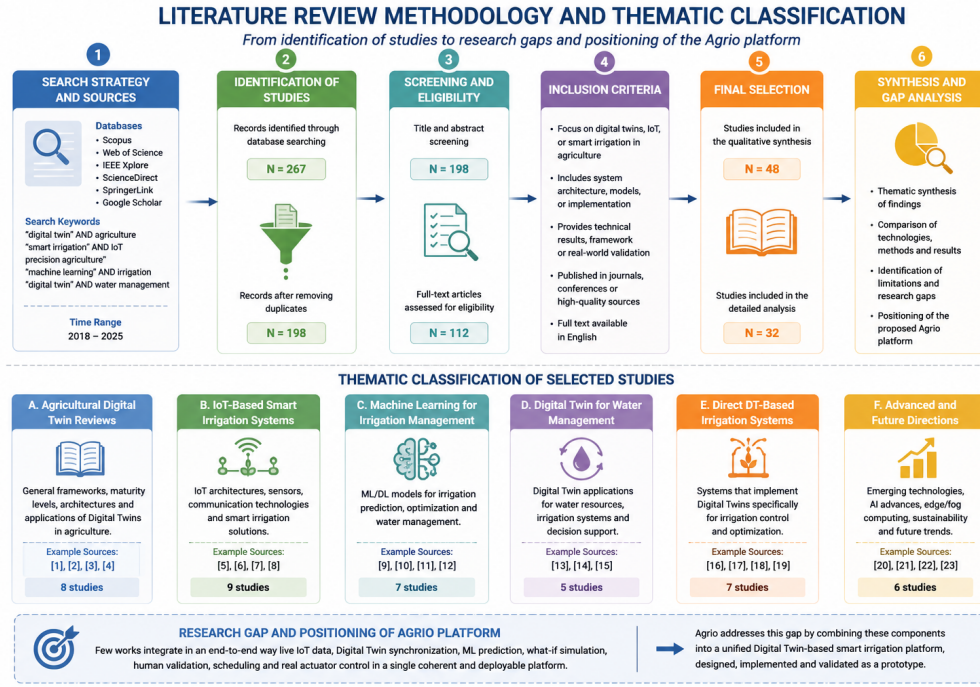


Figure 3.1: Literature review methodology and thematic classification. Source: Authors.

3.2 General agricultural Digital Twin reviews

Subeesh and Chauhan [1] provide a broad review of Agricultural Digital Twins. They describe ADTs as virtual representations that support decision-making, monitoring, simulation, and optimization in smart farming. Their review emphasizes enabling technologies such as IoT, AI, AR/XR, edge computing, federated learning, and sensors. A key point is that agriculture is still at an early stage of Digital Twin adoption, and that practical implementation is complicated by the stochastic nature of weather, soil, and biological processes.

Zhang et al. [2] present a comprehensive review of Digital Twin technology in agriculture. They discuss the distinction between Digital Model, Digital Shadow, and Digital Twin, and describe architecture layers such as perception, network, and application layers. This classification is particularly useful for the present thesis because it helps evaluate whether an agricultural system is only a monitoring platform or whether it includes feedback and decision capacity.

Gund et al. [3] use bibliometric analysis to map research trends in smart agricultural Digital Twins. They show that publications have grown rapidly since 2018 and that the main clusters include precision farming, IoT integration, AI, cyber-physical systems, controlled-environment agriculture, sustainability, and food systems. This confirms that Agrio is positioned inside an active and expanding research field.

Awais et al. [4] focus on precision agriculture through Digital Twins and smart farming technologies. They note that current research often remains isolated around specific applications such as irrigation, disease detection, or crop monitoring rather than providing unified, scalable frameworks. This observation supports the need for integrated platforms like Agrio that connect multiple components: sensing, backend, database, ML, simulation, dashboard, and actuation.

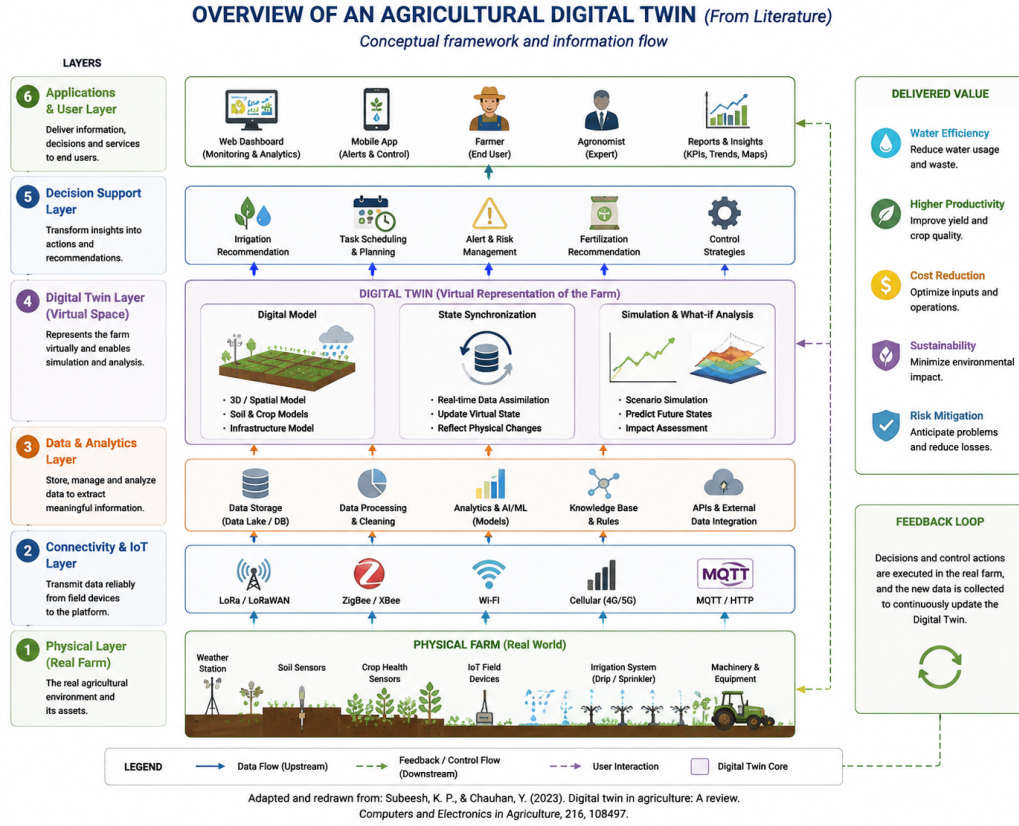


Figure 3.2: Example of agricultural Digital Twin overview from literature [insert or redraw based on Subeesh and Chauhan, Fig. 1]. Source: Adapted/redrawn from [1].

3.3 AI-enabled Digital Twins in agriculture

Tsaousidis et al. [5] provide a recent review focused on AI-enabled Digital Twins in agriculture. They emphasize that many works use the term Digital Twin loosely and sometimes describe systems that are closer to decision support tools than fully synchronized Digital Twins. This warning is important for the scientific rigor of this thesis. The proposed system must therefore explicitly demonstrate its Digital Twin properties: synchronization from sensor data, virtual state, prediction, simulation, recommendation, and feedback/control capability.

The same review highlights data integration as a major bottleneck, especially when combining satellite data, local sensor data, and simulation outputs. Agrio cur-

rently integrates local sensor data, weather/forecast data, and ML outputs, but it does not yet integrate satellite or UAV data. This limitation becomes a clear future direction.

3.4 Digital Twins for agricultural water management

Ahsen et al. [6] specifically review Digital Twins for sustainable agricultural water management. Their review identifies monitoring Digital Twins and predictive Digital Twins. Monitoring Digital Twins focus on real-time response and environmental control, while predictive Digital Twins support proactive management through forecasting and simulation. Agrio combines both aspects: it monitors current zone state and also simulates future moisture evolution.

Digital Twin water management research often focuses on irrigation, hydroponics, aquaponics, and vertical farming. The main benefits include precise resource allocation, scenario planning, predictive maintenance, and optimized water usage. The main barriers include scalability, cost, data integration, and the difficulty of field deployment. These findings align with the scope and limitations of Agrio.

3.5 IoT-based smart irrigation reviews

García et al. [7] review IoT-based smart irrigation systems and analyze sensors, actuators, nodes, wireless technologies, and monitored parameters. They show that smart irrigation systems rely heavily on soil and weather monitoring. Soil moisture is a central variable, but weather parameters such as temperature, humidity, rainfall, wind, and radiation are also important. Their review supports the variable selection used in Agrio.

Obaideen et al. [8] provide an overview of smart irrigation using IoT and emphasize the relationship between irrigation systems and sustainable development. They describe smart irrigation as a system including data acquisition, irrigation control, wireless communication, data processing, and fault detection. Agrio follows these components but extends them with Digital Twin simulation and AI prediction.

Morchid et al. [10] systematically review IoT and ML applications in sustainable irrigation. Their PRISMA-based analysis highlights the benefits of real-time environmental information and ML-based decision-making, while also emphasizing barriers such as complexity, technical constraints, and limited adoption. Their work supports the decision to combine IoT, ML, and human validation rather than relying

on fully autonomous decisions.

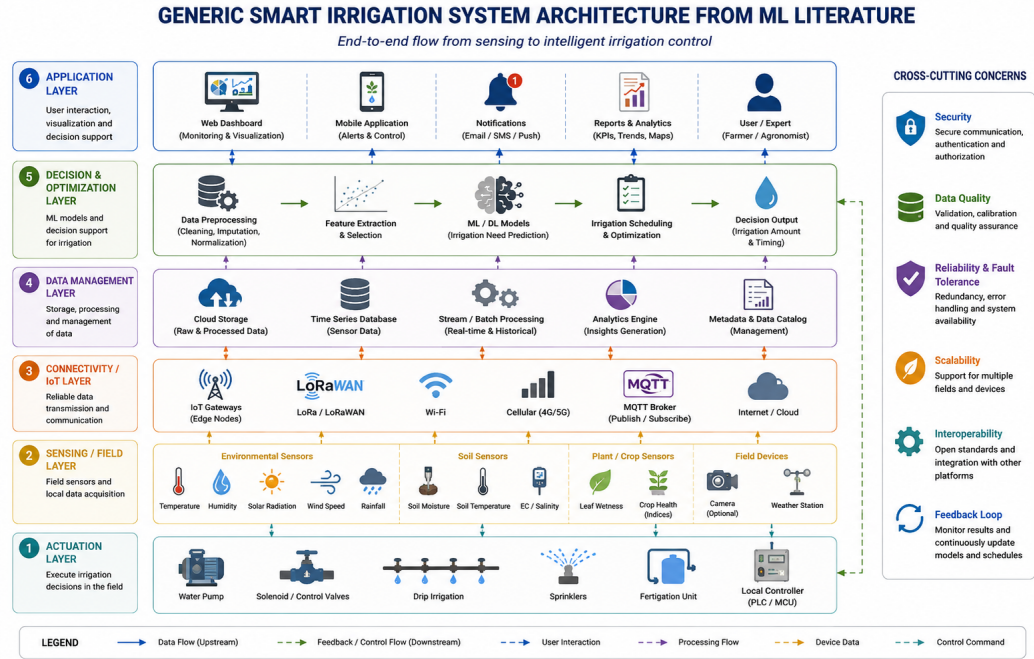


Figure 3.3: Generic smart irrigation system architecture from ML literature [insert/redraw based on Del-Coco et al., Fig. 1]. Source: Adapted/redrawn from [9].

3.6 Machine learning for smart irrigation

Del-Coco et al. [9] focus on machine learning for smart irrigation. They define a generic smart irrigation system as a combination of sensors, IoT technologies, algorithms, datasets, and decision services. They discuss existing ML strategies and note open issues such as long-term data processing and the lack of annotated datasets. This is important for Agrio because the Random Forest models must be presented as part of a prototype and not as universally validated agronomic models.

The reviewed ML literature includes decision trees, Random Forests, SVM, gradient boosting, neural networks, and LSTM models. Random Forests are used in several smart irrigation studies for binary irrigation decisions or pump activation/deactivation. This supports the selection of Random Forest in Agrio, especially because the project uses structured tabular features derived from sensor and environmental data.

One challenge in ML irrigation is interpretability. Farmers may not trust a black-box output if it simply says irrigate or do not irrigate. Agrio addresses this partially by showing confidence, recommended action, suggested time, duration, and explanation in the dashboard. The Digital Twin simulation also provides visual and numerical evidence of expected moisture evolution.

3.7 Directly related Digital Twin irrigation studies

Manocha et al. [15]: IoT-Digital Twin-inspired smart irrigation for optimal water utilization. Uses IoT sensors, virtual counterparts, irrigation requirement calculation, and simulation. Very close to Agrio conceptually; useful for comparing Digital Twin maturity and architecture.

Tancredi et al. [16]: Digital Twin enhanced with ML algorithms for irrigation management using sensor data. Uses ML and sensor data for irrigation ON/OFF decisions, with evaluation metrics such as accuracy, precision, recall, F1, and confusion matrix. Very close to Agrio ML layer.

Qin et al. [17]: Digital Twin-enabled intelligent irrigation-drainage system for water-salt management. Advanced system for precision water-salt management, useful as an example of deeper agronomic modeling and field-oriented Digital Twin irrigation.

Millán et al. [13]: Digital Twin for water-efficient management in a processing tomato commercial farm. Important real-field validation on a 15 ha tomato farm using soil moisture, climate data, simulation models, and daily scheduling. Useful to show what Agrio should target in future field deployment.

Bellvert et al. [14]: Sentinel-2 variables assimilated into a Digital Twin for vineyard irrigation scheduling. Demonstrates remote sensing integration, automated scheduling, and Digital Twin water balance simulation. Useful for future extension with satellite/UAV data.

El Hachimi et al. [19]: Digital Twins and deep reinforcement learning for irrigation water distribution. Advanced optimization direction beyond Agrio, useful for future work.

Liu et al. [18]: Cotton canopy Digital Twin under water stress. Shows plant-water stress modeling and crop visualization aspects, useful for discussing advanced crop-specific twins.

Table 3.1: Key literature themes used in this thesis

Theme	Representative sources	Main insight for Agrio
Agricultural Digital Twins	[1], [2], [3], [4], [5]	DTs require virtual representation, synchronization, simulation, prediction, and feedback; agriculture remains an emerging application domain.

Theme	Representative sources	Main insight for Agrio
Water-management DTs	[6], [13], [14], [15], [17]	Irrigation DTs can support daily scheduling, water balance simulation, and scenario-based management.
IoT smart irrigation	[7], [8], [10], [11], [12]	Smart irrigation systems depend on sensing, wireless communication, cloud processing, and actuator control.
ML irrigation decisions	[9], [10], [16], [21]	ML can classify irrigation need and suggest timing, but requires reliable datasets and clear evaluation.
Advanced future directions	[14], [18], [19]	Remote sensing, water-stress modeling, and reinforcement learning represent future improvements for Agrio.

3.8 Research gaps identified in the literature

The reviewed literature reveals several gaps. First, many agricultural Digital Twin works remain conceptual or focused on monitoring rather than full decision and actuation loops. Second, many smart irrigation systems use threshold rules or simple dashboards without simulation. Third, ML irrigation papers often focus on model accuracy but do not integrate the model into a complete deployed system. Fourth, systems with strong agronomic validation may not provide full software architecture details. Fifth, very few academic prototypes clearly combine physical sensors, MQTT ingestion, database persistence, ML inference, Digital Twin simulation, human validation, web/mobile interfaces, scheduling, and actuator feedback in one coherent workflow.

Agrio addresses part of this gap by implementing an integrated platform. It does not claim to solve every agronomic challenge, but it demonstrates how a Digital Twin irrigation system can be architected and deployed from the physical prototype to the dashboard and pump. Its novelty is therefore mainly architectural and integrative: combining multiple layers that are often studied separately.

3.9 Conclusion

This chapter reviewed the state of the art of Digital Twins in agriculture, IoT-based smart irrigation, ML irrigation decision-making, and direct Digital Twin irrigation applications. The literature shows strong interest in DTs but also highlights fragmentation, limited field validation, and integration challenges. The next chapter compares existing works more systematically and positions Agrio with respect to scientific and engineering criteria.

Part III

Synthesis

Chapter 4

Comparative Analysis and Scientific Positioning

4.1 Comparison criteria

To position Agrio scientifically, existing works are compared according to criteria that reflect Digital Twin maturity and smart irrigation functionality. The criteria include application domain, sensing layer, communication layer, data storage, machine learning, simulation, recommendation, user validation, actuation, deployment, and validation context. These criteria allow a fair comparison between research papers that may focus on different aspects of the problem.

The purpose of this comparison is not to claim that Agrio is superior to field-validated agricultural systems. Instead, it identifies what the project contributes as an integrated engineering prototype and where it remains limited compared with advanced research systems.

Table 4.1: Comparison criteria for Digital Twin irrigation systems

Criterion	Question evaluated
Physical sensing	Does the system receive real sensor data?
Communication	Which technology transmits data?
Data management	Is data stored and structured?
Virtual state	Is there a synchronized virtual representation?
ML/AI	Is prediction or optimization used?
Simulation	Are what-if or future scenarios simulated?

Criterion	Question evaluated
Recommendation	Does the system generate decisions?
Human validation	Can the user validate/override decisions?
Actuation	Can the system control pump/valves?
Deployment	Is there a reproducible implementation?
Validation	Is it tested in lab, prototype, or real farm?

4.2 Comparative matrix

Table 4.2: Scientific comparison of selected works with Agrio

Study	Focus	Main components	DT maturity	Position relative to Agrio
García et al. [7]	IoT smart irrigation review	Sensors, nodes, WSN/IoT, actuators	No specific DT; broad review	Supports Agrio sensing and communication background
Obaideen et al. [8]	IoT smart irrigation overview	Soil/weather monitoring, control, SDG link	No specific DT implementation	Supports sustainability and smart irrigation motivation
Del-Coco et al. [9]	ML smart irrigation review	Physical, processing, dataset, decision layers	No specific prototype	Supports Agrio ML and layered architecture
Kodali & Sarjerao [11]	Low-cost MQTT irrigation	Soil moisture, ESP8266, MQTT, pump status	IoT control, no DT simulation	Similar MQTT/pump logic but simpler than Agrio
Balaceanu et al. [12]	Libelium precision agriculture monitoring	Libelium station, temperature, soil humidity, pressure, radiation	Monitoring platform	Supports Agrio hardware choice

Study	Focus	Main components	DT maturity	Position relative to Agrio
Manocha et al. [15]	IoT-DT smart irrigation	IoT sensors, simulation, irrigation suggestions	Integrated DT framework	Close to Agrio; useful benchmark
Tancredi et al. [16]	DT + ML irrigation	Sensor data, ML classification	DT enhanced with ML	Close ML comparison; Agrio adds full dash-board/scheduler/control
Qin et al. [17]	DT irrigation-drainage	IoT, LSTM, water-salt management	Advanced irrigation DT	Agronomically stronger; Agrio is broader software prototype
Millán et al. [13]	Tomato farm DT	Soil moisture, climate, simulation, daily scheduling	Commercial field DT	Field validation stronger than Agrio; future target
Bellvert et al. [14]	Vineyard DT with Sentinel-2	Remote sensing, fAPAR, SWB simulation, scheduling	Commercial vineyard DT	Shows future remote sensing integration
Agrio	This thesis	Libelium sensors, XBee, Raspberry Pi, MQTT, FastAPI, PostgreSQL, RF ML, simulation, dashboard, WeMos pump	Human-supervised operational prototype DT	Integrative architecture: sensing-to-actuation pipeline tested in controlled prototype

4.3 Positioning of Agrio by Digital Twin maturity

Agrio can be positioned between a Digital Shadow and a fully autonomous Digital Twin. It has a real sensing layer and updates the virtual zone state using live MQTT data, which goes beyond an offline Digital Model. It also generates decisions and can control the pump through MQTT commands, which goes beyond a one-way Digital Shadow. However, because autonomous actuation is intentionally limited and the prototype is not yet validated in a real farm over a full growing season, the system should be described as a human-supervised operational Digital Twin prototype.

This positioning is academically honest and defensible. It acknowledges that field-validated systems such as Irri_DesK applied to commercial tomato or vineyard plots have stronger agronomic validation. At the same time, it highlights Agrio’s engineering completeness: backend architecture, database model, ML service, simulation interface, web/mobile dashboard, scheduling, and actuator feedback.

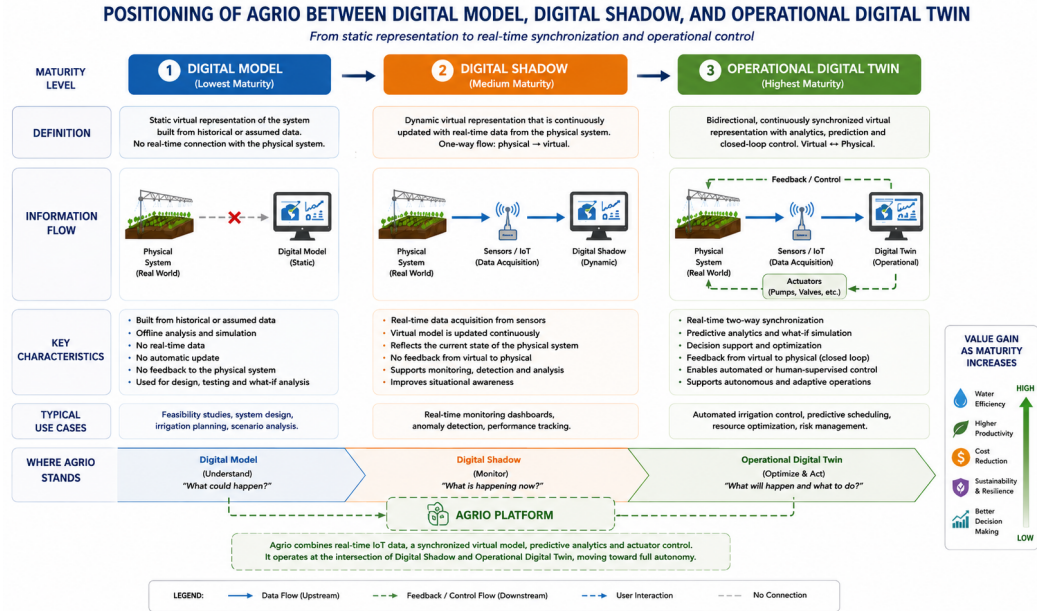


Figure 4.1: Positioning of Agrio between Digital Model, Digital Shadow, and operational Digital Twin. Source: Authors.

4.4 Novelty of the proposed project

Table 4.3: Novelty and contribution of Agrio

Contribution	Explanation
End-to-end integration	Many works focus on one layer; Agrio connects field sensing, RF communication, gateway, MQTT, backend, database, ML, simulation, dashboard, scheduling, and pump control.
Physical prototype	The system is not only conceptual; it uses a controlled prototype with Libelium kit, XBee, Raspberry Pi, WeMos, and water pump.

Contribution	Explanation
Human-supervised Digital Twin	Recommendations are not blindly executed. The user can validate, schedule, or manually control the pump.
What-if simulation	The platform compares no-irrigation, with-irrigation, and ML-selected paths before physical action.
Deployment-oriented engineering	Docker Compose and separated services make the prototype reproducible and easier to demonstrate.
Traceable database model	The schema stores measurements, states, history, predictions, simulations, recommendations, commands, events, and feedback.

4.5 Weaknesses and limitations identified from the comparison

The comparison also reveals weaknesses. Agrio is currently validated in a controlled prototype environment, not in a full commercial farm. The ML models rely on adapted datasets and project data, but not on multi-season field datasets. Agronomic calibration of crop coefficients, soil hydraulic parameters, and water balance models is still limited. The system does not yet include remote sensing, UAV imagery, salinity management, reinforcement learning, or optimization of multi-zone water distribution.

Another limitation is security. A production irrigation platform would require robust authentication, authorization, encrypted device provisioning, secure MQTT credentials, audit logs, and protection against unauthorized pump commands. The current project prioritizes functional integration for academic validation, while production security remains future work.

These limitations do not reduce the value of the prototype; rather, they define a realistic boundary. The thesis should present Agrio as a strong final-year engineering prototype and research-informed Digital Twin platform, not as a commercially validated agricultural product.

4.6 Conclusion

This chapter positioned Agrio among existing smart irrigation and Digital Twin systems. The contribution lies in integrating multiple layers into an operational prototype and demonstrating the Digital Twin loop from sensing to simulation and control. The comparison also clarified the limitations of the project, especially the need for real field validation, stronger agronomic calibration, and production-level security. The next part presents the requirements and architecture of the proposed system.

General Conclusion

This Master Diploma report presented the scientific study of Agrio, a Digital Twin-based intelligent irrigation platform for precision agriculture. The work focused on the background, state of the art, and synthesis required to understand how Digital Twins can support smart irrigation and agricultural water management.

The background chapters showed that smart irrigation is motivated by water scarcity, climate variability, and the need to improve irrigation decisions. Precision agriculture, IoT, wireless communication, artificial intelligence, simulation, and human-in-the-loop decision support form the main technological foundation of this field. Digital Twin technology extends classical monitoring by connecting the physical state of an agricultural system with a virtual representation able to support prediction, simulation, recommendation, and feedback.

The state-of-the-art review showed that Digital Twins are increasingly studied in agriculture but remain difficult to deploy due to data heterogeneity, biological complexity, cost, connectivity, calibration, and limited long-term validation. The literature also shows that many systems focus on monitoring or decision support but do not always implement a complete sensing-to-actuation loop.

The synthesis and comparative analysis positioned Agrio as an integrated project that combines concepts from smart irrigation, agricultural Digital Twins, IoT, machine learning, simulation, human validation, and actuation. This positioning does not claim that Agrio is a fully mature field-deployed Digital Twin. Instead, it identifies the scientific gap that the engineering prototype addresses: building a coherent bridge between research concepts and an operational cyber-physical irrigation workflow.

Future scientific work should extend the review with new field-validated systems, compare additional machine learning and optimization approaches, study agronomic calibration in greater depth, and evaluate Digital Twin maturity using long-term agricultural deployments.

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